ALGEBRA 2 · UNIT 2: LINEAR FUNCTIONS & SYSTEMS

Farm Planner

Use linear programming and systems in three variables to run a farm.

CP2

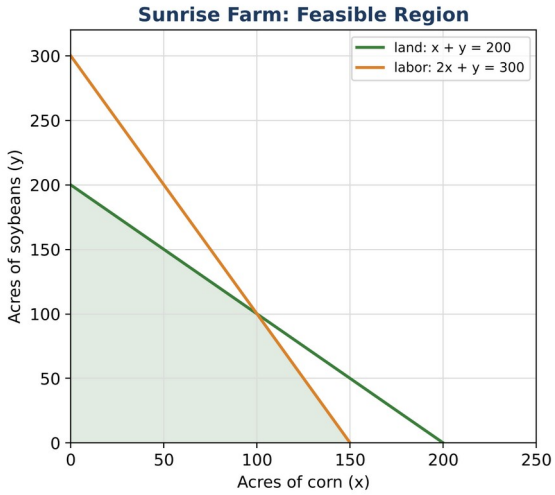
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Sunrise Farm grows corn and soybeans. Land and labor are limited, and each crop brings a different profit. The farmer wants the plan that earns the most money.

This is linear programming: graph the constraints, find the corners of the feasible region, and test them. Then you will mix feed for the farm's animals with a three-variable system.



Per Acre
Corn: 2 h labor
profit $300
Soybeans: 1 h labor
profit $200
Land ≤ 200 acres
Labor ≤ 300 h

Learning Goals

- I can write and graph constraints as inequalities.
- I can find the vertices of a feasible region.
- I can use vertices to find the maximum profit.

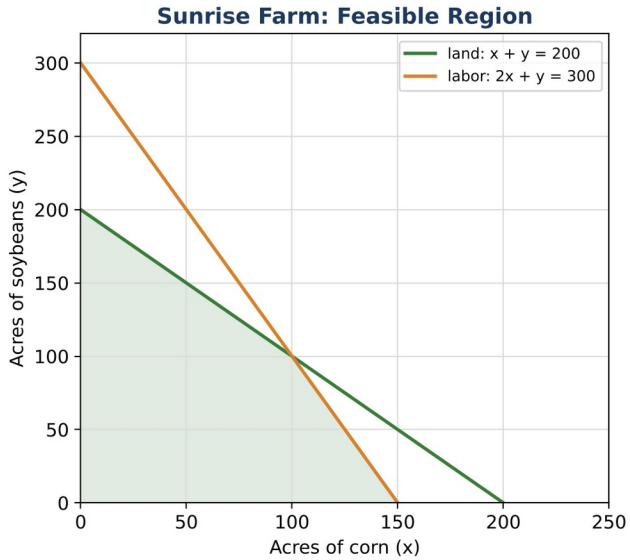
Project Timeline

When	What you will do
Day 1	Constraints
Day 2	Find the Best Plan
Day 3	Feed & Plan
Final	Farm Business Plan + reflection

Part 1: Constraints

Day 1 · Let x = acres of corn and y = acres of soybeans.

1. The farm has at most 200 acres. Write an inequality.



Per Acre
Corn: 2 h labor
profit $300
Soybeans: 1 h labor
profit $200
Land ≤ 200 acres
Labor ≤ 300 h

2. Corn needs 2 hours of labor per acre and soybeans need 1 hour. There are at most 300 hours. Write an inequality.

3. Why must x ≥ 0 and y ≥ 0?

4. Is the plan (90, 100) possible? Is (120, 90)? Check both constraints.

Part 2: Find the Best Plan

Day 2 · Profit P = 300x + 200y.

5. Find the vertex where the land and labor lines meet by solving the system x + y = 200 and 2x + y = 300.

6. List all four vertices of the feasible region.

7. Find the profit at each vertex. Which plan earns the most?

Vertex	Profit 300x + 200y
(0, 0)	
(0, 200)	
(150, 0)	

Part 3: Feed & Plan

Day 3 · Feed mix: corn costs $0.10 per lb and soy meal costs $0.30 per lb.

8. The farmer needs 100 lb of feed that costs $16 total. Write and solve a system for the pounds of corn and soy meal.

9. Write a short business plan stating how many acres of each crop to plant and why.

10. What would change if soybean profit rose to $350 per acre?

Final Product: Farm Business Plan

Your final product must include:

- Graph of constraints with the feasible region shaded
- Vertex table with profits
- Feed mix solution
- Business plan

Reflection

R1. Why is the best plan always at a corner?

R2. What does a point outside the region mean for the farm?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Recommendation is the true optimum (or a tie is explained), and the feed system is solved and checked. Goes beyond the minimum with a creative, realistic choice.	Recommendation is the true optimum (or a tie is explained), and the feed system is solved and checked.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Business plan includes the constraints, graph, vertices, optimal plan, and the feed mix solution. Polished, organized, and visually engaging.	Business plan includes the constraints, graph, vertices, optimal plan, and the feed mix solution. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.